

TUSHAR BHARADWAJ

Spatial Data Scientist | Remote Sensing & Geospatial ML
ttushar684@gmail.com | +91 98828 78776 | Bangalore, India

29 March 2026

Hiring Committee

CIFOR-ICRAF — SPACIAL Unit
Nairobi, Kenya (Remote Position)

Re: Spatial Data Scientist – Remote Sensing

Dear Hiring Committee,

I am applying for the Spatial Data Scientist position within CIFOR-ICRAF's SPACIAL unit. Over the past six and a half years I have worked almost exclusively on geospatial ML, SAR-optical fusion, and satellite-based ecosystem monitoring, so the work this role describes is not unfamiliar territory for me. What draws me specifically to SPACIAL is the monitoring challenge in dryland greening landscapes. In areas targeted by the Great Green Wall with persistent cloud cover and sparse ground truth make optical-only analysis brittle. That is precisely where SAR time series and LSTM-based trend models can provide the kind of analytical continuity that programme teams actually need.

At Thryve Earth, I build CNN and LSTM pipelines for forest cover change, biomass estimation, and LULC classification using Sentinel-1, Sentinel-2, and Landsat 8/9 data, reaching 85 to 90 percent accuracy across multi-million hectare areas. Beyond the modelling, I designed and delivered a full-stack decision-support platform from scratch using React, Python, and GCP. It handles everything from site eligibility screening and suitability mapping to carbon modeling and farmer onboarding, and it is live across five districts in Karnataka and three in Indonesia. The users range from field farmers to investors, which has taught me a lot about making spatial outputs genuinely usable outside a technical audience.

My two years with Plant-for-the-Planet and IIT Roorkee are where my research practice took shape. The LSTM vegetation trend forecasting model I built on Sentinel-2 VI time series is work I am actively preparing for peer-reviewed publication. Applying to CIFOR-ICRAF is a conscious decision to move from applied delivery into an institution where that kind of methodological contribution can be properly validated and built upon. I have also written funding proposals for IKI large grants and TFFF compliance programmes, so I understand what it takes to sustain research at scale.

I want to be upfront about the degree requirement. My postgraduate background is an M.Tech in engineering rather than a PhD or MSc in remote sensing. What I have is a PG Diploma from IIIT Bangalore in data science, an advanced certification from IIT Roorkee in GIS and remote sensing, and six years of applied research output at ecosystem scale. I believe that is a substantive equivalency, even if it is a non-traditional one. I am committed to formalising this through publication and would appreciate the chance to discuss how CIFOR-ICRAF thinks about practitioner-researcher profiles.

If given the opportunity, I would bring a production-ready SAR–optical workflow combined with advanced deep learning, designed for challenging landscapes like those of the Great Green Wall. I also offer platform delivery experience to translate these outputs into practical tools for field teams and decision-makers. I would be glad to talk through how that fits what SPACIAL is working on.

Yours sincerely,

Tushar Bharadwaj